

Classification of Elements & Periodicity

A. Historical Development of Periodic Classification

1. Döbereiner's Law of Triads had limited success mainly because it was:
 - (A) applicable only to a few sets of elements
 - (B) based on atomic masses that were unknown at the time
 - (C) valid only for elements of the same valency
 - (D) in violation of the octet rule
2. Newlands' Law of Octaves states that when elements are arranged in order of increasing atomic mass, every eighth element has properties similar to the:
 - (A) first element
 - (B) second element
 - (C) fourth element
 - (D) sixth element
3. Mendeleev's periodic law states that the properties of elements are a periodic function of their:
 - (A) atomic number
 - (B) atomic mass
 - (C) number of valence electrons
 - (D) number of protons and neutrons
4. Moseley's work on X-ray spectra led to the Modern Periodic Law, which states that the properties of elements are a periodic function of their:
 - (A) atomic mass
 - (B) atomic number
 - (C) atomic volume
 - (D) atomic radius
5. Mendeleev left a gap in his periodic table for an undiscovered element he called "eka-silicon"; this element was later identified as:
 - (A) germanium
 - (B) gallium
 - (C) scandium
 - (D) technetium

B. The Long Form of the Periodic Table

6. The number of periods and groups in the long form of the modern periodic table are, respectively:
 - (A) 7 and 18
 - (B) 8 and 16
 - (C) 7 and 8
 - (D) 6 and 18
7. An element with the electronic configuration $[\text{Ar}] 3d^6 4s^2$ belongs to which block of the periodic table?
 - (A) s-block
 - (B) p-block
 - (C) d-block
 - (D) f-block

8. The elements of Group 17 in the periodic table are collectively known as:

- (A) alkali metals
- (B) alkaline earth metals
- (C) halogens
- (D) noble gases

9. Which period of the periodic table contains only two elements?

- (A) Period 1
- (B) Period 2
- (C) Period 3
- (D) Period 4

10. The lanthanides and actinides together constitute the:

- (A) s-block
- (B) p-block
- (C) d-block
- (D) f-block

C. IUPAC Nomenclature of Elements

11. The IUPAC name for the element with atomic number 118 is:

- (A) Ununoctium
- (B) Oganesson
- (C) Livermorium
- (D) Tennessine

12. In IUPAC nomenclature of elements, the digit "1" is represented by the root:

- (A) nil
- (B) un
- (C) bi
- (D) tri

13. The IUPAC symbol for the element with atomic number 115 (Moscovium) is:

- (A) Uup
- (B) Mc
- (C) Fl
- (D) Lv

D. Atomic and Ionic Radii

14. On moving from left to right across a period, atomic radius generally:

- (A) increases
- (B) decreases
- (C) remains constant
- (D) first increases, then decreases

15. On moving down a group, atomic radius generally:

- (A) increases
- (B) decreases
- (C) remains constant
- (D) shows no regular trend

16. Among the isoelectronic species Na^+ , Mg^{2+} and Al^{3+} , the correct order of decreasing ionic radius is:
- (A) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 - (B) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$
 - (C) $\text{Mg}^{2+} > \text{Na}^+ > \text{Al}^{3+}$
 - (D) $\text{Al}^{3+} > \text{Na}^+ > \text{Mg}^{2+}$
17. The correct order of increasing atomic radii for C, N, O and F is:
- (A) $\text{F} < \text{O} < \text{N} < \text{C}$
 - (B) $\text{C} < \text{N} < \text{O} < \text{F}$
 - (C) $\text{F} < \text{N} < \text{O} < \text{C}$
 - (D) $\text{O} < \text{N} < \text{F} < \text{C}$
18. A cation is always smaller than its parent atom mainly because:
- (A) the nuclear charge decreases on losing an electron
 - (B) the number of occupied shells often decreases and the effective nuclear charge per remaining electron increases
 - (C) the electron cloud expands
 - (D) protons are removed along with electrons
19. An anion is always larger than its parent atom mainly because:
- (A) electron–electron repulsion increases while the nuclear charge stays the same
 - (B) the nuclear charge increases
 - (C) a new energy shell is always added
 - (D) the number of protons increases

E. Ionisation Enthalpy

20. Ionisation enthalpy generally increases across a period from left to right because:
- (A) atomic size increases
 - (B) effective nuclear charge increases and atomic size decreases
 - (C) the shielding effect increases
 - (D) the number of shells increases
21. Ionisation enthalpy generally decreases down a group because:
- (A) nuclear charge decreases
 - (B) atomic size and the shielding effect increase
 - (C) electron affinity decreases
 - (D) electronegativity increases
22. The first ionisation enthalpy of boron is lower than that of beryllium because:
- (A) boron has a larger atomic size than beryllium
 - (B) the 2p electron in boron is more easily removed than an electron from beryllium's stable, filled $2s^2$ subshell
 - (C) beryllium has a lower nuclear charge
 - (D) boron has a fully filled 2p subshell
23. The first ionisation enthalpy of nitrogen is higher than that of oxygen because:
- (A) nitrogen has a smaller atomic size
 - (B) nitrogen has an extra-stable, half-filled $2p^3$ configuration
 - (C) oxygen has a lower nuclear charge
 - (D) nitrogen has a higher electron affinity

24. The second ionisation enthalpy of an element is always _____ its first ionisation enthalpy.
- (A) less than
 - (B) greater than
 - (C) equal to
 - (D) unrelated to
25. Among Na, Mg and Al, the element with the lowest first ionisation enthalpy is:
- (A) Na
 - (B) Mg
 - (C) Al
 - (D) all three are equal
26. The first ionisation enthalpies of Li, Be and B follow the irregular order $\text{Be} > \text{B}$. This is attributed to:
- (A) increasing atomic size from Li to B
 - (B) the extra stability of the completely filled $2s^2$ configuration in Be
 - (C) decreasing nuclear charge from Li to B
 - (D) poor shielding by the 1s electrons
27. Among Na, Mg, Al and Cl, the element with the highest first ionisation enthalpy is:
- (A) Na
 - (B) Mg
 - (C) Al
 - (D) Cl

F. Electron Gain Enthalpy

28. Electron gain enthalpy generally becomes more negative across a period from left to right because:
- (A) atomic size increases
 - (B) effective nuclear charge increases, making it easier to attract an added electron
 - (C) the shielding effect increases
 - (D) electropositive character increases
29. Electron gain enthalpy generally becomes less negative down a group because:
- (A) atomic size increases, weakening the attraction for an added electron
 - (B) nuclear charge increases
 - (C) ionisation enthalpy decreases
 - (D) electronegativity increases
30. Although fluorine is smaller than chlorine, chlorine has a more negative electron gain enthalpy than fluorine because:
- (A) fluorine has a higher nuclear charge
 - (B) fluorine's very small, compact 2p subshell causes strong electron–electron repulsion that partly offsets the attraction for an incoming electron
 - (C) chlorine has a lower electronegativity
 - (D) fluorine is a liquid at room temperature
31. Noble gases have a large positive electron gain enthalpy mainly because:
- (A) they have a very small atomic size
 - (B) their stable, completely filled configuration resists the addition of an extra electron
 - (C) they have low ionisation enthalpy
 - (D) they have high electronegativity

32. Among the halogens, the element with the most negative electron gain enthalpy is:

- (A) F
- (B) Cl
- (C) Br
- (D) I

33. The correct order of increasing negative electron gain enthalpy for N, O and F is:

- (A) $N < O < F$
- (B) $F < O < N$
- (C) $O < N < F$
- (D) $F < N < O$

G. Electronegativity

34. Electronegativity generally increases across a period and decreases down a group. The most electronegative element in the periodic table is:

- (A) oxygen
- (B) chlorine
- (C) fluorine
- (D) nitrogen

35. On the Pauling scale, electronegativity is a measure of:

- (A) the energy released when an isolated atom gains an electron
- (B) the tendency of a bonded atom to attract the shared pair of electrons in a covalent bond
- (C) the energy required to remove an electron from a gaseous atom
- (D) the size of an atom

36. The correct order of increasing electronegativity for C, N, O and F is:

- (A) $C < N < O < F$
- (B) $F < O < N < C$
- (C) $C < F < N < O$
- (D) $N < C < O < F$

37. Lithium shows a close similarity in electronegativity and other properties to which Group 2 element, an example of the diagonal relationship?

- (A) Magnesium
- (B) Calcium
- (C) Strontium
- (D) Beryllium

38. As the electronegativity of an element increases, its metallic character generally:

- (A) increases
- (B) decreases
- (C) remains unaffected
- (D) first increases, then decreases

H. Metallic Character & Nature of Oxides

39. Metallic character generally _____ across a period and _____ down a group.
- (A) increases; increases
 - (B) decreases; increases
 - (C) increases; decreases
 - (D) decreases; decreases
40. Oxides of elements on the left side of the periodic table (metals) are generally:
- (A) acidic
 - (B) basic
 - (C) neutral
 - (D) amphoteric
41. Oxides of the non-metals on the right side of a period are generally:
- (A) acidic
 - (B) basic
 - (C) neutral
 - (D) amphoteric
42. Which of the following oxides is amphoteric in nature?
- (A) Na_2O
 - (B) Al_2O_3
 - (C) SO_3
 - (D) MgO
43. Which of the following oxides is the most acidic?
- (A) Na_2O
 - (B) MgO
 - (C) Al_2O_3
 - (D) Cl_2O_7
44. Which of the following represents a basic oxide?
- (A) CO_2
 - (B) K_2O
 - (C) P_2O_5
 - (D) SO_2

I. Diagonal Relationship & Periodic Anomalies

45. Lithium shows a diagonal relationship with which element, owing to a similar charge/size ratio?
- (A) Sodium
 - (B) Magnesium
 - (C) Beryllium
 - (D) Aluminium
46. Beryllium shows a diagonal relationship with:
- (A) Boron
 - (B) Aluminium
 - (C) Silicon
 - (D) Magnesium

47. The anomalous behaviour of the first element of each group, compared with the other members of that group, is mainly attributed to its:

- (A) unusually high electronegativity and small size
- (B) low ionisation enthalpy
- (C) large atomic radius
- (D) low nuclear charge

48. Which of the following statements about the diagonal relationship in the periodic table is correct?

- (A) It occurs because diagonally placed elements have comparable size and similar charge density
- (B) It applies uniformly to all elements in the periodic table
- (C) It occurs only among the noble gases
- (D) It results from identical electronic configurations

J. Miscellaneous Periodic Trends

49. Which of the following statements about elemental abundance is correct?

- (A) Oxygen is the most abundant element in the Earth's crust
- (B) Hydrogen is the most abundant element in the Earth's crust
- (C) Silicon is the most abundant element in the universe
- (D) Helium is the most abundant element in the Earth's crust

50. Mercury is unusual among metals in that it:

- (A) has the highest melting point of all metals
- (B) exists as a liquid at room temperature
- (C) is the most chemically reactive metal
- (D) has the smallest atomic radius among metals

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	B	31	B	41	A
2	A	12	B	22	B	32	B	42	B
3	B	13	B	23	B	33	A	43	D
4	B	14	B	24	B	34	C	44	B
5	A	15	A	25	A	35	B	45	B
6	A	16	A	26	B	36	A	46	B
7	C	17	A	27	D	37	A	47	A
8	C	18	B	28	B	38	B	48	A
9	A	19	A	29	A	39	B	49	A
10	D	20	B	30	B	40	B	50	B